

MACHINE TOOLS

Introduction

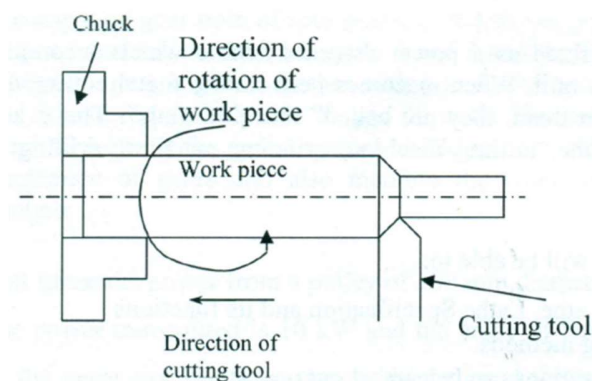
A machine tool may be defined as a power driven machine which accomplishes the cutting or machining operations on it. When machines perform the metal cutting operations using the cutting tools mounted on them, they are called "machine tools". The examples of such machine tools are Lathe, Drilling Machine, Milling Machine, Grinding Machine, Shaping Machine, Planing Machine etc.

Lathe

Lathe is a machine tool employed generally to produce circular objects. The main function of the lathe is to remove metal from a piece of work to give it the required shape and size. This is accomplished by holding the work securely and rigidly on the machine using a work holding device and then turning it against the cutting tool which will remove the metal from the workpiece in the form of chips. To cut the material properly the tool should be harder than the material of the workpiece. Lathe is so versatile that almost all the operations that can be performed using other machine tools like drilling machine, milling machine, shaping machine etc. can also be performed on it.

Principle of working of a Lathe:

Lathe works on the principle that a cutting tool can remove material in the form of chips from rotating work pieces to produce circular objects. It is illustrated in the figure below.

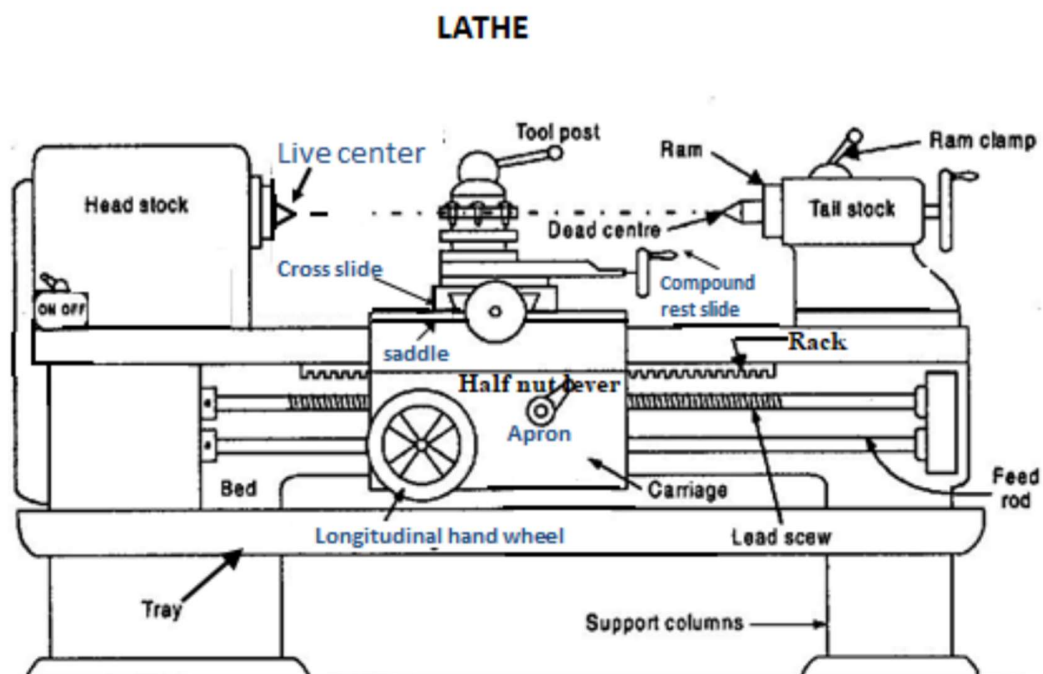


Principle of Working of a Lathe

The work piece is held rigidly using one of the work holding devices, known as chuck and is rotated at high speeds in an anticlockwise direction when viewed from the tailstock end or the right side end. A V - shaped cutting tool which is stronger and harder than the work piece is held against the workpiece and is fed or moved parallel to the lathe axis towards the headstock end or the left side end leading to the production of a cylindrical surface.

Major parts of a Lathe and their functions

The figure below illustrates the major parts of a lathe.



Bed: The bed is a foundation part of a lathe and supports all its other parts. There are two precision machined guide ways on the top of the bed viz., outer guide ways and inner guide ways. The headstock and tailstock are mounted on the inner guideways and the carriage assembly is supported by the outer guideways.

Headstock: The housing comprising the stepped cone pulley and feed gear box is called the headstock of the lathe. The headstock is rigidly mounted on the inner guide ways of lathe bed at

its left end and is not movable. The main spindle projects out from the headstock and is driven at multiple speeds. Headstock is also called live centre since it is power driven.

Tailstock: Tailstock is the movable part of the lathe that carries the dead centre in it. It is mounted loosely on the inner guide ways of lathe bed on the right hand side and can be moved and locked in any position. It supports the free end of the work piece. Tailstock is also used to clamp tools like twist drills and reamers for making holes, taps and dies for cutting threads.

Carriage Assembly: The carriage assembly serves to support the tool and rides over the outer guide ways longitudinally between headstock and tailstock. It is composed of 5 main parts.

- i. **Saddle:** Saddle is a H shaped casting that slides over the outer guide ways and serves as the base for the cross slide, compound rest and tool post.
- ii. **Cross slide:** Cross slide is mounted on the saddle and enables the movement of the cutting tool in a lateral direction (perpendicular to the movement of the saddle) across the lathe bed either automatically or manually by means of cross feed hand wheel. It supports the compound rest and the tool post.
- iii. **Compound Rest:** Compound rest is mounted on the top of the cross slide and supports the tool post. It has a circular base graduated in degrees and can be swiveled to any angle in the horizontal plane to facilitate the taper turning operation. There is no power feed to the compound rest. It is moved manually by the compound rest feed handle independent of the lathe cross feed.
- iv. **Apron:** Apron is mounted at the front of the saddle beneath it and houses carriage and cross slide mechanisms. The apron hand wheel moves carriage manually by means of the rack and pinion arrangement.
- v. **Tool Post:** Tool Post is mounted in the T - slot of the compound rest. The tool post clamps the tool holder in the proper position for carrying out various machining operations.

Lead Screw: Lead screw is a screw rod which runs longitudinally in front of the lathe bed. The rotation of the lead screw moves the carriage to and fro longitudinally during thread cutting

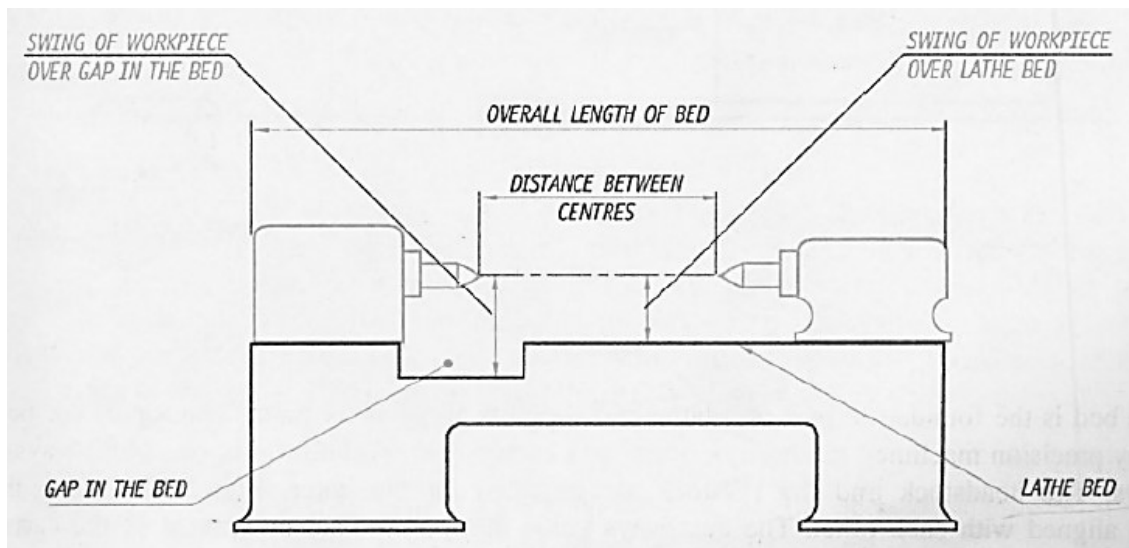
operation. It may also be used to provide motion for turning operations in lathes that are not equipped with a feed rod.

Feed rod: The feed rod is a long shaft mounted in front of the lathe bed. The power is transmitted from the lathe spindle to the apron gears through a feed rod via large number of gears. It facilitates automatic longitudinal movement of the carriage during turning and lateral movement of the cross slide during facing operations. It is not used for thread cutting operation.

Main drive: It consists of an electric motor mounted in the lower left portion of the lathe in conjunction with the transmission system like belt or gear drive thereby driving the cone pulley. Various spindle speeds can be obtained by shifting the belt on different steps of the cone pulley. Spindle speeds can be further varied using the back gear arrangement by engaging or disengaging the same.

Lathe Specification:

The size of the lathe is specified by the following as shown in the figure below.



Specification of a Lathe

1. Distance between centers: It is the maximum distance that can be obtained between the lathe centers. It represents the maximum length of the work piece that can be held between centers.
2. Overall length of the bed: It is the maximum length of the bed starting from the headstock end to tailstock end.
3. Swing of workpiece over lathe bed: It is the largest diameter of the work piece that can be rotated without touching the bed.
4. Swing of workpiece over gap in the bed: It is the maximum diameter and the width of a work piece that can swing when the lathe has gap in the bed.

Classification of Lathe:

Lathes are classified as:

1. Bench Lathe
2. Speed Lathe
3. Engine Lathe
4. Capstan and Turret Lathe
5. Tool Room Lathe
6. Automatic Lathe
7. CNC Lathe

1. Bench Lathe: This lathe is the smallest in the lathe family. It derives its name from the fact that it is mounted on a bench. It has all the parts of a conventional lathe and is generally used for small and precision related work.

2. Speed Lathe: It consists of a bed, a headstock, a tailstock and a tool post mounted on an adjustable slide. There is no feed box, lead screw or conventional type of carriage. The tool is mounted on an adjustable slide and is fed by hand control. The headstock construction is very simple and has only two or three spindle speeds available. It derives its name because of the very high speed of the headstock spindle. It is used in operations such as woodworking, polishing, spinning etc.

3. Engine Lathe: This lathe is the most important and widely used one in the lathe family. It derives its name owing to the fact that early lathes were driven by steam engines. It has all the basic parts and the headstock is much more robust in construction and has additional mechanisms for driving the lathe spindle at multiple speeds. It can feed the cutting tool both in

cross and longitudinal directions with reference to the lathe axis with the help of carriage, feed rod and lead screw. It is a versatile machine adapted for every type of lathe work.

4. Capstan and Turret Lathe: This lathe is an development over an engine lathe and is used for mass production work. The distinguishing feature of this type of lathe is that the tailstock is replaced by a hexagonal turret on the face of which multiple tools can be fitted and fed into the work in proper sequence. The main advantage of this lathe is that several different type of operations can be performed on a workpiece without resetting the work or the tools, and a number of identical parts can be produced in a minimum time.

5. Tool Room Lathe: A tool room lathe has features similar to an engine lathe but is more accurately built and has wide range of spindle speeds ranging from very low to high speeds. This lathe is mainly used for precision work on tools, dies, gauges and in machining work where accuracy is needed.

6. Automatic Lathe: These are high speed, heavy duty, mass production lathes with complete automatic control. Once the tools are set they automatically perform all the operations to finish the job. The changing of tools, speed and feeds are also done automatically. Using this lathe identical parts can be produced in large numbers.

7. CNC Lathe: A CNC lathe is an improvement over an automatic lathe by incorporating a computer interface. It uses a dedicated stored program to carry out different operations. High degree of precision and accuracy can be achieved by carrying out different operations on a workpiece. The program can be suitably changed to meet the versatile requirements. It is used for mass production work.

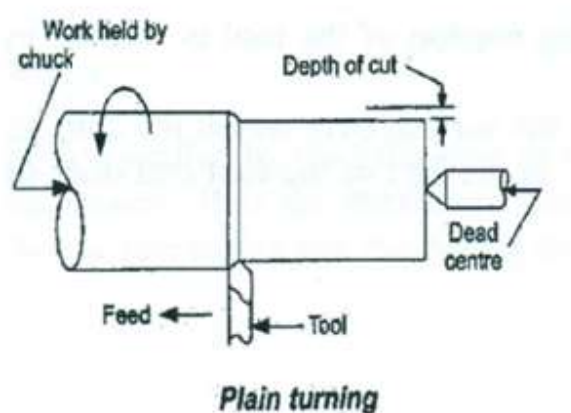
Lathe Operations:

All most all the basic machining operations can be performed on a lathe. Some of the important and generally performed lathe operations are

1. Turning
2. Step Turning
3. Facing.
4. Taper turning

- 5. Thread cutting
- 6. Knurling.
- 7. Chamfering.

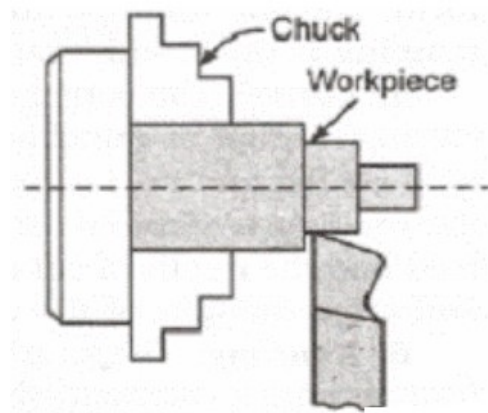
1. Turning or Plain Turning: The process of metal removal from a cylindrical job is called straight or plain turning. The plain turning operation is illustrated in the figure below.



Plain Turning

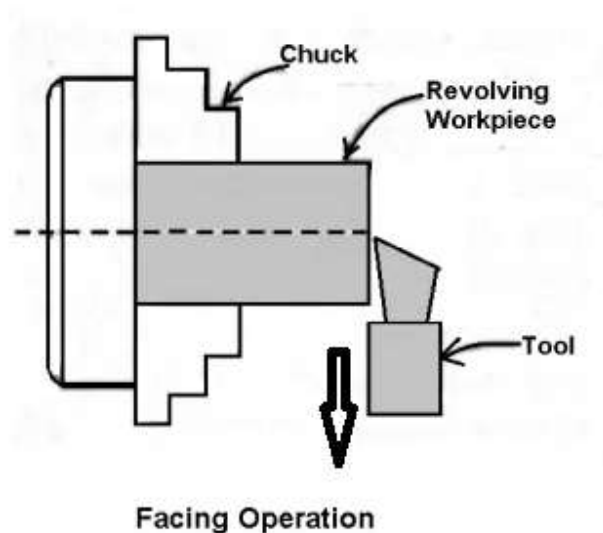
Cross slide and the carriage are used to perform the plain turning operation. The workpiece is supported between the two centres which cause the rotation of the workpiece. A single point cutting tool is fed perpendicular to the axis of the workpiece with the required depth of cut using the cross slide hand wheel and is moved parallel to the axis of the workpiece towards the headstock using the carriage hand wheel facilitating the metal removal. Plain turning operation is generally performed in two steps namely Rough turning and Finish turning. In rough turning excess material is removed from the workpiece using a rough turning tool in a minimum time by applying a high rate of feed and heavy depth of cut. In finish turning a smooth surface is generated by using a finish turning tool by applying high cutting speed, small feed and very small depth of cut.

2. Step turning: The step turning operation is illustrated in the figure below.



Step turning is an operation of producing multiple steps of different diameters on the same cylindrical specimen. The work is held in between the lathe centers and the depth of cut to obtain the step on the cylinder is provided by cross slide movement and feed by the carriage movement.

3. Facing: Facing is an operation of producing a flat surface on the ends of the work piece perpendicular to the axis. This can also be used to reduce the length of the work piece so as to conform to the required length. It is illustrated in the figure below.



To perform the operation the tool tip is set at the same height as that of the work piece axis and is fed gradually by moving the cross slide in a direction perpendicular to the axis of the work piece

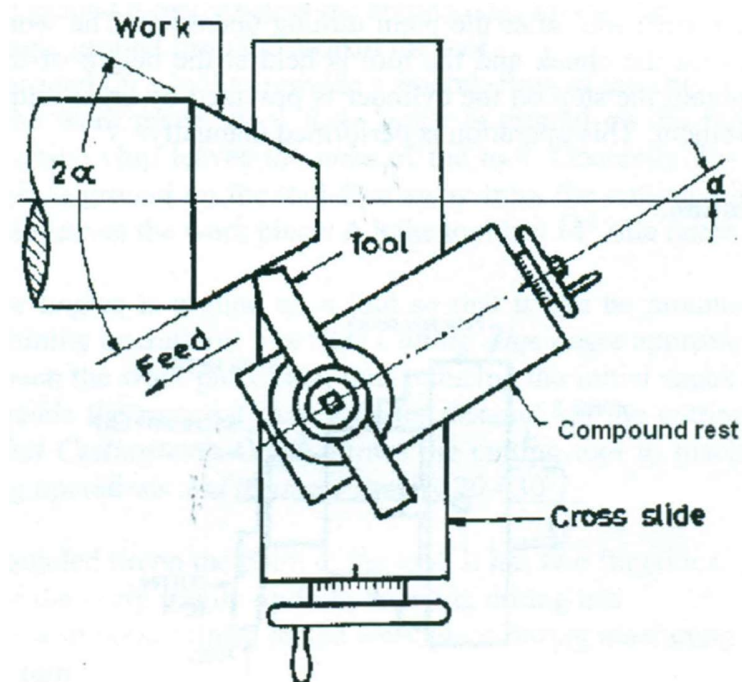
generally from the centre progressing towards the outer edge of the workpiece. For large and rough surfaces facing also can be performed in roughing and finishing cycles.

4. Taper Turning: Taper turning is an operation of producing a conical surface on a cylindrical specimen. It is achieved by gradually reducing the diameter from a cylindrical surface. The tapering of a part has wide applications in the construction of machines. Almost all machine spindles have taper holes which receive taper shanks of various tools and work holding devices. The operation can be carried out either by rotating the job normally and feeding the tool at some angle or by rotating the job at an offset angle and feeding the tool normally. The widely used methods of taper turning are

1. Taper turning by swiveling the compound rest and
2. Taper turning by tailstock set over method.

1. Taper Turning by Swiveling Compound Rest Method:

The taper turning operation by swiveling the compound rest is illustrated in the figure below.



Taper turning by swiveling the compound rest

This method employs the principle of turning taper by rotating the workpiece on the lathe axis and feeding the tool at an angle to the axis of rotation of the workpiece. The tool mounted on the compound rest has a circular base graduated in degrees, which may be swiveled and clamped at

any desired angle. Once the compound rest is set at the desired half taper angle, rotation of the compound slide wheel will cause the tool to be fed at that angle and generate the corresponding taper. This method is limited to turn a short taper owing to the limited movement of the compound rest. The movement of the tool in this method is purely controlled by hand.

The taper angle by which the compound rest is to be rotated is given by the relation.

$$\tan \alpha = \frac{D-d}{2L}$$

where,

α = Half taper angle (simply taper angle) in degrees.

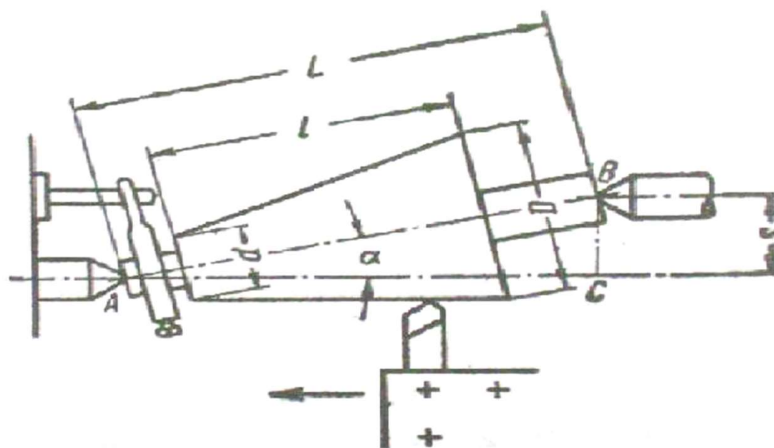
D = Larger diameter of the taper in mm.

d = Smaller diameter of the taper in mm.

L = Length of the taper in mm.

2. Taper turning by Tail Stock Set over Method (Offset method):

This method of taper turning is illustrated in the figure below.



Taper turning by tailstock set over method

The principle of turning taper by this method is to shift the axis of rotation of the workpiece at an angle to the lathe axis, and feeding the tool parallel to the lathe axis. The angle at which the axis of rotation of the workpiece is shifted is equal to half angle of the taper. This is done by sliding the body of the tailstock on its base towards or away from the operator using the set over screws situated on either side at the base of the tailstock and a scale attached to the base of the tailstock. In this method the axis of the work piece gets inclined at an angle with the longitudinal movement of the tool which will be parallel to the lathe bed. Entire carriage has to be moved parallel to the lathe bed to cut the taper. Since the amount set over being limited this method is suitable for turning small taper on long jobs.

In this method the amount of set over required to machine a particular taper is given by the relation

$$\text{Set Over} = \frac{L(D-d)}{2l}$$

Where L= Length of the work.

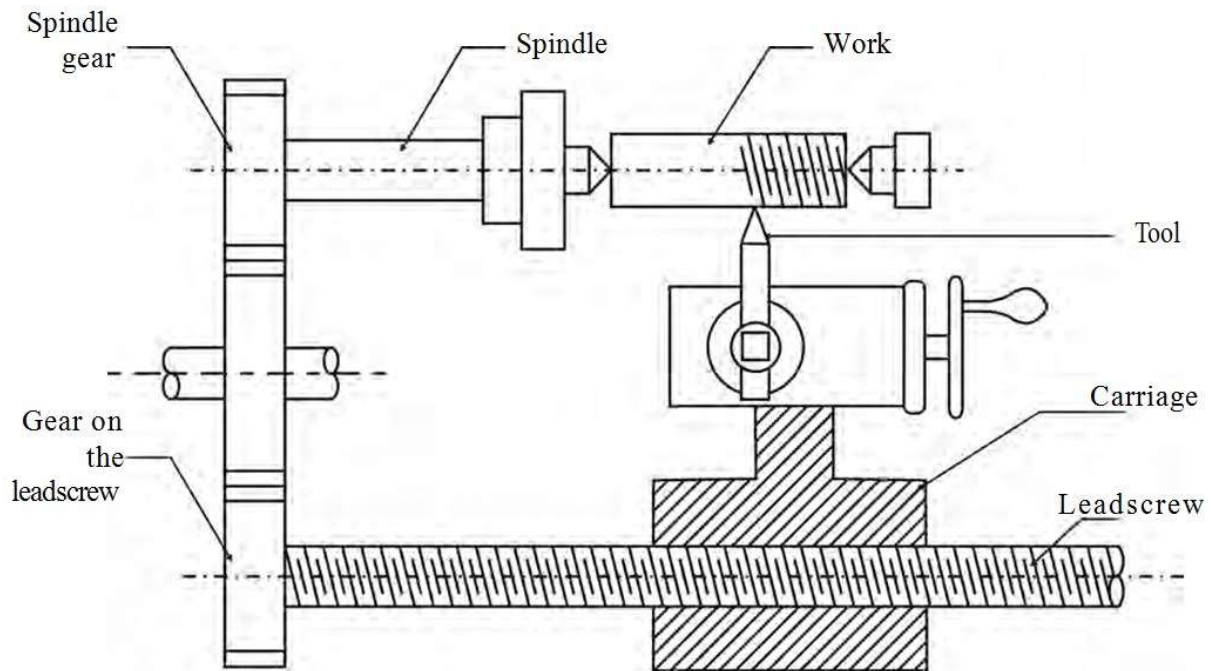
l= Length of the taper.

D= Large diameter of the taper.

d= Small diameter of the taper.

5. Thread Cutting:

Thread cutting is one of the most important operations performed on a lathe. It is an operation of producing a helical groove of specific shape say V or square on a cylindrical surface. The operation involves feeding the tool longitudinally when the job is revolved between the centres or by a chuck. In thread cutting the longitudinal feed should be equal to the pitch of the thread to be cut per revolution of the workpiece. In other words the tool must travel by a distance equal to the pitch of the thread per revolution of the workpiece. The lead screw of the lathe through which the saddle receives its traversing or longitudinal motion when the half nut lever is engaged has a definite pitch. A definite ratio should therefore be found out between the longitudinal feed and rotation of the workpiece or headstock spindle so that the relative speeds of rotation of the leadscrew and the work will result in the cutting of a screw of desired pitch. This is effected by the change gears arranged between the spindle and the leadscrew or by a change gear mechanism or feed box in modern lathes. The thread cutting operation is illustrated in the figure below.

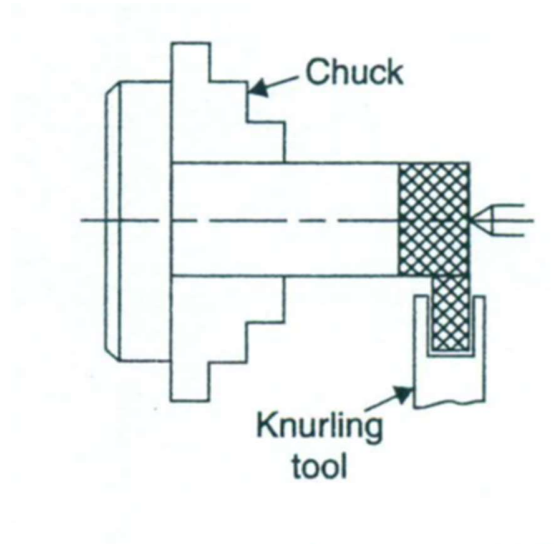


Thread cutting

In thread cutting operation the first step is to reduce the diameter of the cylindrical specimen equal to the major diameter of the screw thread. Change gears of correct size are then fitted at the left end of the bed between the spindle and the leadscrew. Initially the tool is brought to the starting point of the job and a small depth of cut is given to the tool using the cross slide. The lathe is switched on and the carriage is engaged with the lead screw using the half nut lever which enables the tool to give the first cut. After the tool has produced the helical groove up to the end of the work it is quickly withdrawn using the cross slide, the half nut is disengaged and the tool is brought back to the starting position to give a fresh cut. Since several cuts are necessary before the full depth of the thread is reached, before reengaging the half nut for the next cut it is necessary to ensure that the tool will follow the same path it has traversed in the previous cut, otherwise the job will be spoiled. This is ensured by a attachment called thread chasing dial.

For cutting right hand threads the carriage must move towards the headstock and for left hand threads it must move towards the tailstock. The job must always rotate in the anticlockwise direction when viewed from the tailstock end.

6. Knurling:

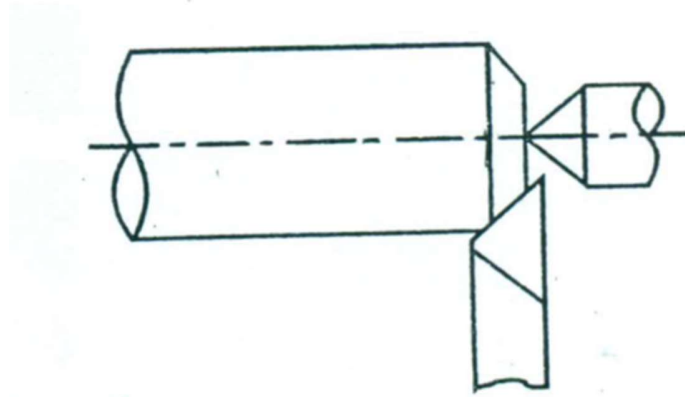


Knurling Operation

Knurling is a slow speed operation of generating a serrated surface on a cylindrical workpiece or it is the process of embossing a diamond shaped impression pattern on the surface of a cylindrical workpiece using a special knurling tool. The purpose of knurling is to provide an effective gripping surface on a workpiece to prevent it from slipping when operated or held by hand.

The knurling tool consists of a set of hardened steel rollers in a holder with the teeth cut on their surface in a definite pattern. While carrying out knurling operation the back gear is engaged and the tool is held rigidly on the tool post and the rollers are pressed against the revolving work piece to squeeze the metal against the multiple cutting edges thereby producing depressions in a regular pattern on the surface of the workpiece.

7. Chamfering:



Chamfering Operation

Chamfering is an operation of beveling the extreme ends of a workpiece. This is done to remove the burrs, to protect the ends of the workpiece from getting damaged and to have a better look. The operation may be performed after knurling, turning, thread cutting etc. It is an essential operation after thread cutting so that the nut may pass freely on the threaded workpiece.

Drilling:

Drilling is an operation of producing holes in a work piece using a drill bit. The stationary work is held in a fixture and the rotating tool is fed against the workpiece vertically downwards. Drilling Machine is a power operated machine tool, which holds the drilling tool known as twist drill in its rotating spindle and produces a hole when moved linearly against the work piece. The twist drill having sharp cutting edges is provided with a helical groove along its length to allow for the chip or the cut material to escape through it. A liquid coolant is generally used while drilling to remove the heat of friction and to obtain a better finish for the hole.

Drilling machines may be classified as

1. Portable Drilling Machine.
2. Bench or Sensitive Drilling Machine.
3. Upright Drilling Machine.
4. Gang Drilling Machine.

5. Multiple Spindle Drilling Machine

6. Radial Drilling Machine.

1. Portable Drilling Machine:

A portable drilling machine is generally employed for drilling holes in light classes of work such as structural applications, fitting work in assemblies and in cases where high accuracy is not required. These are small in size, compact and carry a small electric motor either A.C. or D.C. inside them. They can drill holes upto 12 mm in diameter. They run at very high speeds as they are required to produce small holes for shorter depths.

2. Bench or Sensitive Drilling Machine

Bench or Sensitive drilling machines are light duty drilling machines widely used in general workshops. They are fitted on a workbench and consists of a vertical column, a horizontal table, a head supporting the motor and a driving mechanism and a vertical spindle for driving and rotating the drill. In this machine there is no arrangement for automatic feed of the drill spindle. The drill is fed purely by hand control and the progress of the drill into the work is sensed by the operator, hence the name sensitive drilling machine. Generally holes upto 15mm in diameter can be drilled using these machines.

3. Upright or Pillar Drilling Machine:

Upright or Pillar drilling machines are designed for handling medium and heavy sized workpieces and can drill holes up to 50mm in diameter. Though in construction this machine is similar to a sensitive drilling machine, but is larger and heavier in comparison to it. It consists of a robust pillar column erected over a sturdy base fixed on the floor which supports an adjustable work table and a drill head assembly. The table may be moved up and down on the column for accommodating workpieces of different heights. It can also be moved in an arc up to 180° around the column and can be clamped in any position. Heavier and odd sized work may be supported directly on the base of the machine and drilled after the arm is swung out of the way. The machine has power feed mechanism and large number of spindle speeds and feeds are available for drilling different types of work.

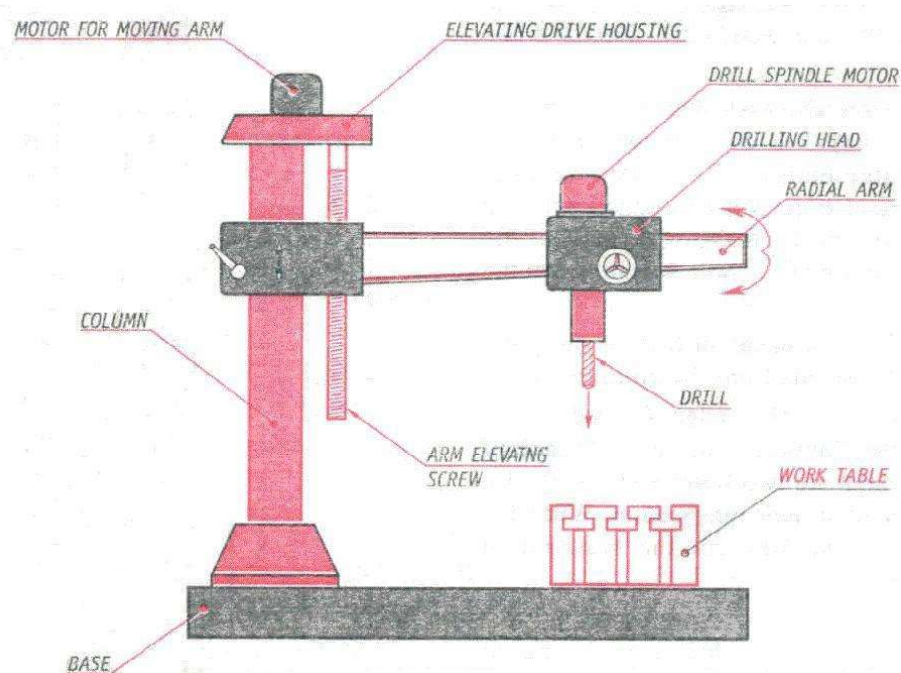
4. Gang Drilling Machine: When a number of single spindle drilling machine columns are placed side by side on a common base and have a common work table, the machine is known as gang

drilling machine. In a gang drilling machine four to six spindle columns may be mounted side by side. In some machines it may be possible to adjust the position of the columns so that the space between the spindles may be varied. The speed and feed of the spindles are controlled independently. This type of machine is generally used for mass production work. A series of operations such as drilling, reaming, countersinking, tapping etc. may be performed on the work simply by shifting the work from one position to other on the work table. Each spindle may be set up properly with different tools for different operations.

5. Multiple Spindle Drilling Machine:

The function of the multiple spindle drilling machine is to drill a number of holes in a piece of work simultaneously and to reproduce the same pattern of holes in a number of identical pieces in mass production work. Such machines have several spindles driven by a powerful single motor and all the spindle holding drills are fed into the work simultaneously. The spindles are so constructed that their centre distance may be adjusted in any position as required by the various jobs within the capacity of the drill head.

6. Radial Drilling Machine:



Radial Drilling Machine

Radial drilling machine is used for medium and heavy duty applications. The construction of a radial drilling machine is illustrated in the figure above. It consists of a heavy,

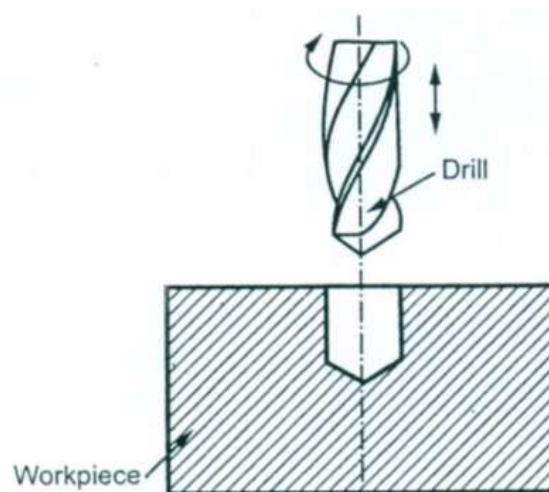
circular column mounted on a very strong base. The column supports a radial arm which can be raised and lowered to accommodate workpieces of different heights. The arm may be swung around to any position over the work bed. The drill head containing mechanism for rotating and feeding the drill is mounted on the radial arm and can be moved horizontally on the guide ways and can be clamped in any desired position. These three movements of the machine permit the drill to be located at any desired position for carrying out the drilling operation. When several holes are to be drilled on a larger workpiece the arm and the drill head may be adjusted so that the drill spindle may be moved from one position to the other without altering the setting of the work. The work may be mounted on the table or when it is very large it may be placed on the floor.

Drilling Machine Operations:

The different operations that can be performed by a drilling machine are as follows:

1. Drilling
2. Reaming
3. Boring.
4. Counter Boring
5. Counter sinking
6. Spot facing
7. Tapping.

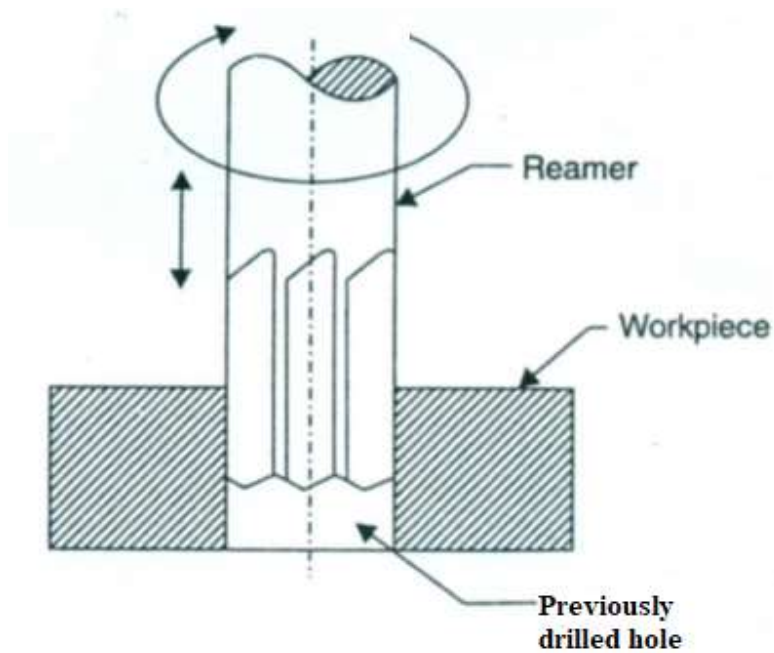
1. Drilling:



Drilling Operation

Drilling is an operation of producing cylindrical holes in a workpiece by removing the metal by the rotating edge of the cutting tool known as the drill. The centre of the hole is located on the workpiece and a indentation is made using the centre punch. The drill bit is then fed at this centre point to produce the required hole. Drilling operation does not produce an accurately sized hole and the surface generated is rough.

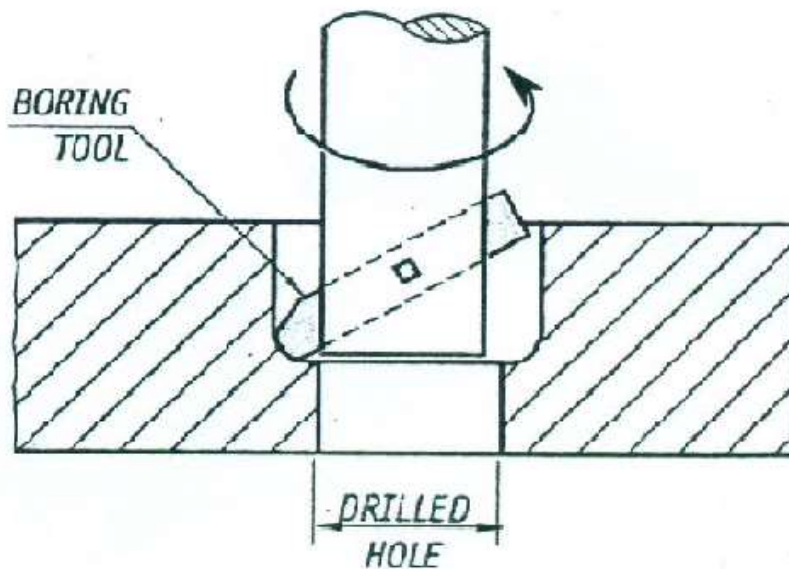
2. Reaming:



Reaming Operation

Reaming is an operation of rightsizing and finishing the previously drilled hole using a tool called reamer. In drilling operation the drilled hole is slightly undersize and has a rough surface. Hence a tool called reamer which has straight multiple cutting edges is mounted in place of the twist drill and reaming is done in the same way as drilling. It removes only a small amount of material and produces a smooth finish on the drilled surfaces. Reamer simply follows the path which has been previously drilled and cannot correct the hole location.

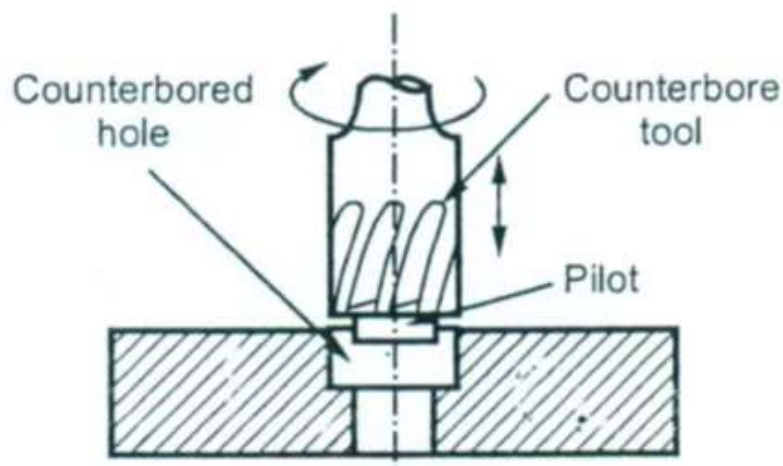
3. Boring:



Boring

Boring is an operation of increasing the size of the already drilled hole. When a suitable size drill is not available, initially a hole is drilled to the nearest size and using a single point cutting tool, the size of the hole is increased to the required size as shown in the figure. By lowering the tool while it is continuously rotating, the size of the hole is increased over its entire depth.

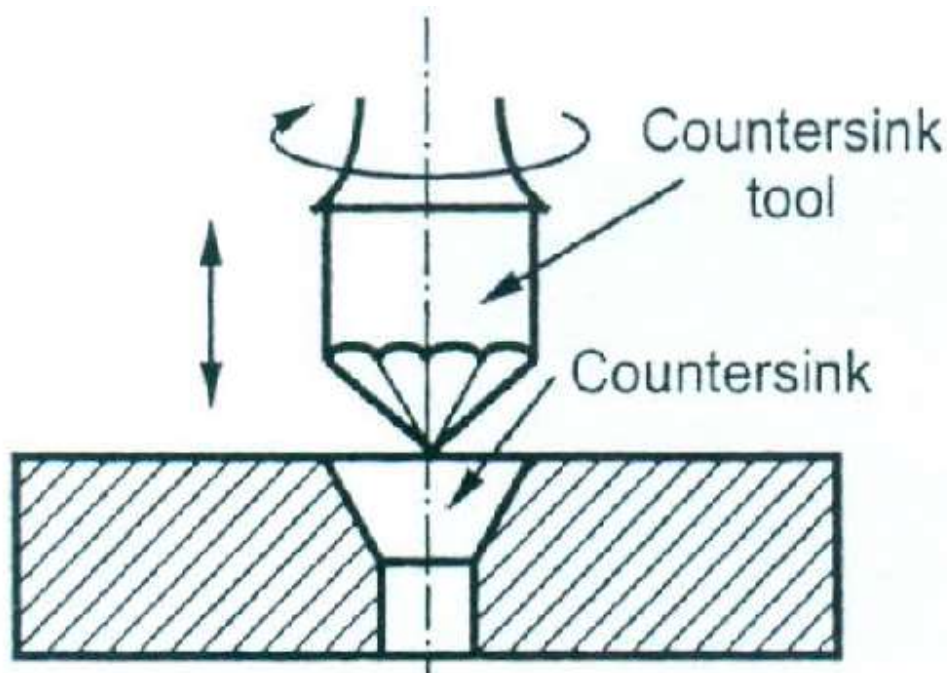
4. Counterboring:



Counterboring

Counterboring is an operation of increasing the size of a hole only at one end through a small depth as shown in the figure. This operation forms a larger sized recess or a shoulder at one end of the existing hole. The cutting tool will have a small cylindrical projection known as pilot to guide the tool while counterboring. The diameter of the pilot will always be equal to the diameter of the previously drilled hole. Counterboring is done on the holes to accommodate the socket head screws, or grooved nuts, or round head bolts.

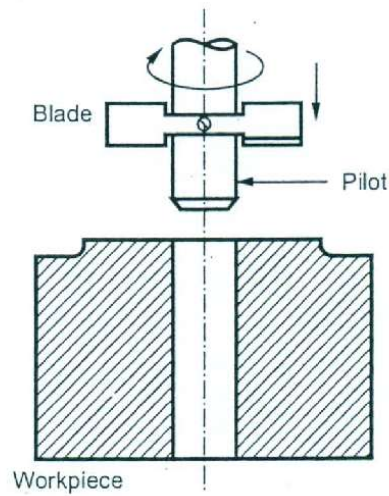
5. Countersinking:



Countersinking

Countersinking is an operation of making the end of the hole into a conical shape. It is done using a countersinking tool as shown in the figure. The countersunk holes are used when the countersunk screws are to be screwed into the holes so that their top faces have to be in flush with the top surface of the work piece.

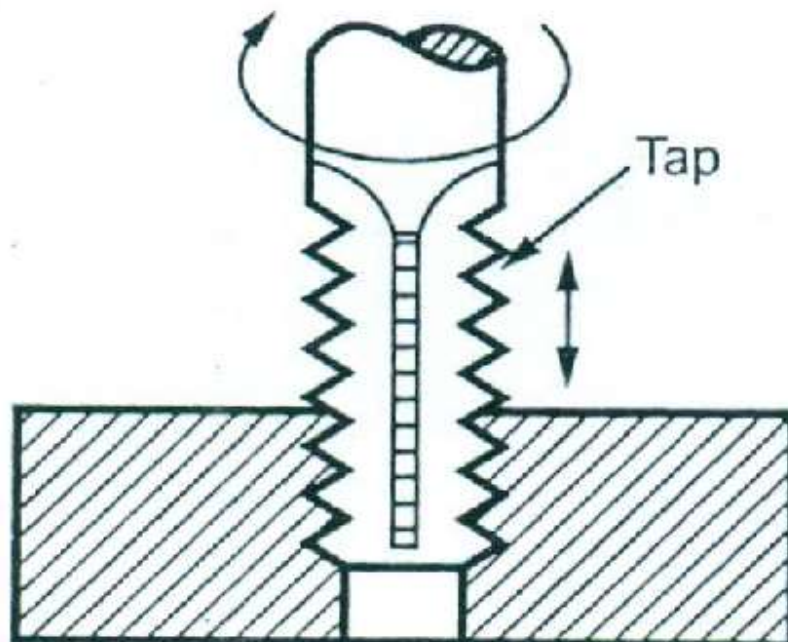
6. Spot facing:



Spotfacing

Spot facing is an operation of producing a flat round surface around a drilled hole to so as to provide a good bearing surface for the proper seating of a bolt head or a nut. This is carried out using a spot facing tool as shown in the figure.

7. Tapping:



Tapping

Tapping is an operation of cutting internal threads using a cutting tool called tap. Initially a hole is drilled which must be smaller than the tap size by twice the depth of the thread. The drill spindle is then fitted with the tap and the feeding is done by operating the feed lever similar to the conventional drilling operation. During this operation the spindle speed should be much lower than that used for conventional drilling operation.